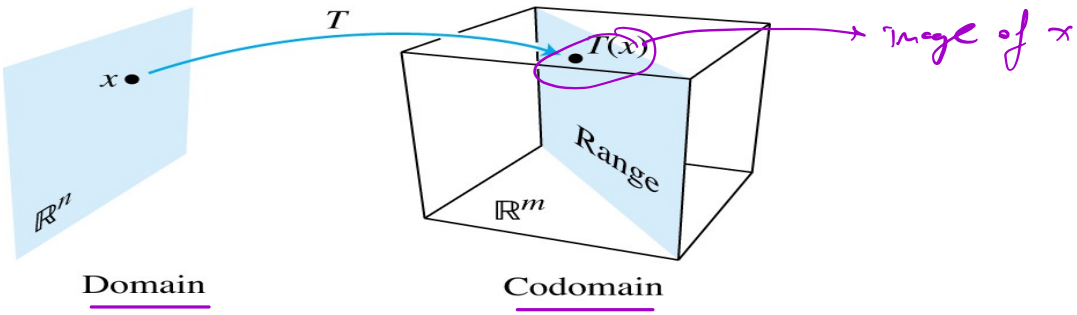
1.8 Linear Transformations

- A **transformation** (or **function** or **mapping**) T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns to each vector $\mathbf{x}$ in $\mathbb{R}^n$ a vector $T(\mathbf{x})$ in $\mathbb{R}^m$.
- The set $\mathbb{R}^n$ is called **domain** of T, and $\mathbb{R}^m$ is called the **codomain** of T.
- The notation $T: \mathbb{R}^n \longrightarrow \mathbb{R}^m$ indicates that the domain of T is $\mathbb{R}^n$ and the codomain is $\mathbb{R}^m$.
- For $\mathbf{x}$ in $\mathbb{R}^n$, the vector $T(\mathbf{x})$ in $\mathbb{R}^m$ is called the **image** of $\mathbf{x}$ (under the action of T).
- The set of all images $T(\mathbf{x})$ is called the **range** of T. See the next figure.



Domain, codomain, and range of $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$.

$A_{m\times n} x_{n\times 1} = T(x)_{m\times 1}$

Matrix Transformations

In the remaining of this section we focus on mappings associated with matrix multiplication.

- For each $\mathbf{x}$ in $\mathbb{R}^n$, $T(\mathbf{x})$ is computed as $A\mathbf{x}$, where A is an mxn matrix.
- For simplicity, we denote such a *matrix transformation* by $x \longmapsto Ax$.
- The domain of T is $\mathbb{R}^n$ when A has n columns and the codomain of T is $\mathbb{R}^m$ when each column of A has m entries.
- The range of T is the set of all linear combinations of the columns of A, because each image $T(\mathbf{x})$ is of the form $A\mathbf{x}$.

Ex:

Let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$, $\mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$, and

define a transformation $T : \mathbb{R}^2 \to \mathbb{R}^3$ by $T(\mathbf{x}) = A\mathbf{x}$, so that

$$T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}$$

a. Find $T(\mathbf{u})$, the image of $\mathbf{u}$ under the transformation T.

b. Find an $\mathbf{x}$ in $\mathbb{R}^2$ whose image under T is $\mathbf{b}$.

c. Is there more than one $\mathbf{x}$ whose image under T is $\mathbf{b}$?

d. Determine if $\mathbf{c}$ is in the range of the transformation T.

$Ax = b$ → unique?

$Ax = c$ → ? → Existence.

a. Compute $T(u) = Au = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1.2 + -3.-1 \\ 3.2 + 5.-1 \\ -1.2 + 7.-1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -5 \end{bmatrix}$

$u \in \mathbb{R}^2$, $T(u) \in \mathbb{R}^3$

b. Solve $T(x) = b$ for x. That is, solve $Ax = b$ s.t

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$$

A x b

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1.5 \\ 0 & 1 & -0.5 \\ 0 & 0 & 0 \end{bmatrix}$$

Then $x_1 = 1.5$, $x_2 = -0.5$, $x = \begin{bmatrix} 1.5 \\ -0.5 \end{bmatrix}$

$T(x) = Ax = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix} = b$

c. It is clear that $Ax = b$ has a unique solution. Just check the reduced echelon form and observe that there are no free variables. Therefore, there is exactly one x whose image is b under T.

d. $T(x) = Ax = c$. This is just another way of asking if the system $Ax = c$ consistent.

Row reduce the augmented matrix:

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -35 \end{bmatrix}$$

echelon form.

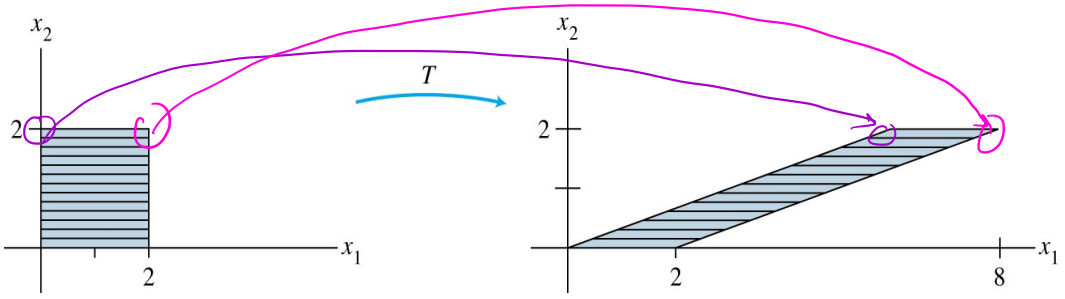
The third equation $0 = -35$, shows that the system is inconsistent. So c is not in the range of T.

Ex:

Let $A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$.

The transformation $T: \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ defined by $T(x) = Ax$ is called a **shear transformation**.

- It can be shown that if T acts on each point in the 2x2 square shown in the next figure, then the set of images forms the shaded parallelogram.



- The key idea is to show that T maps line segments onto line segments and then to check that the corners of the square map onto the vertices of the parallelogram.

For instance, the image of the point $\mathbf{u} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$ is

$T(\mathbf{u}) = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$, and the image of $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ is $\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$.

Definition:

A transformation (or mapping) T is **linear** if:

(i) $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in the domain of T;

(ii) $T(c\mathbf{u}) = cT(\mathbf{u})$ for all scalars c and all $\mathbf{u}$ in the domain of T.

- Linear transformations *preserve the operations of vector addition and scalar multiplication*.
- Property (i) says that the result $T(\boldsymbol{u}+\boldsymbol{v})$ of first adding $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ and then applying T is the same as first applying T to $\mathbf{u}$ and $\mathbf{v}$ and then adding T ($\mathbf{u}$) and T ($\mathbf{v}$) in $\mathbb{R}^m$.
- These two properties lead to the following useful facts.

If T is a linear transformation, then

$$T(\mathbf{0}) = \mathbf{0} \tag{3}$$

and

$$T(c\mathbf{u}+d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v}) \tag{4}$$

for all vectors $\mathbf{u}, \mathbf{v}$ in the domain of T and all scalars c, d.

- Property (3) follows from condition (ii) in the definition, because $T(\mathbf{0}) = T(\mathbf{0}\boldsymbol{u}) = 0T(\boldsymbol{u}) = \mathbf{0}$.
- Property (4) requires both (i) and (ii): $T(c\boldsymbol{u}+d\boldsymbol{v}) = T(c\boldsymbol{u}) + T(d\boldsymbol{v}) = cT(\boldsymbol{u}) + dT(\boldsymbol{v})$.